

ING AI Guild Meetup

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Why Spring AI?

Gen AI - not a Data-science but an Integration problem!

AI has to reach the systems of records - backend services

... services such as Spring Boot services

spring AI

Models

OpenAI
Azure
stability.ai
Hugging Face
ANTHROP\C
MISTRAL AI_
Gemini Google
Amazon Bedrock
MINIMAX

Chat Client

Chat Model

- Multimodality
- Tool Calling
- Struct. Out.
- Prompt

Advisors
Memory

Embedding Model
Image Model
Audio Model
Moderation Model

Decision Models (SystemOne, Jev.)

VectorStore

Pinecone
redis
Weaviate
qdrant
Milvus
elasticsearch
mongoDB.
sandra
Azure
SAP HANA
Chroma

ETL
Modular RAG
...

Model Context Protocol (MCP)
Spring AI Session 2.1+
Observability
Spring AI Agent (2.1+)

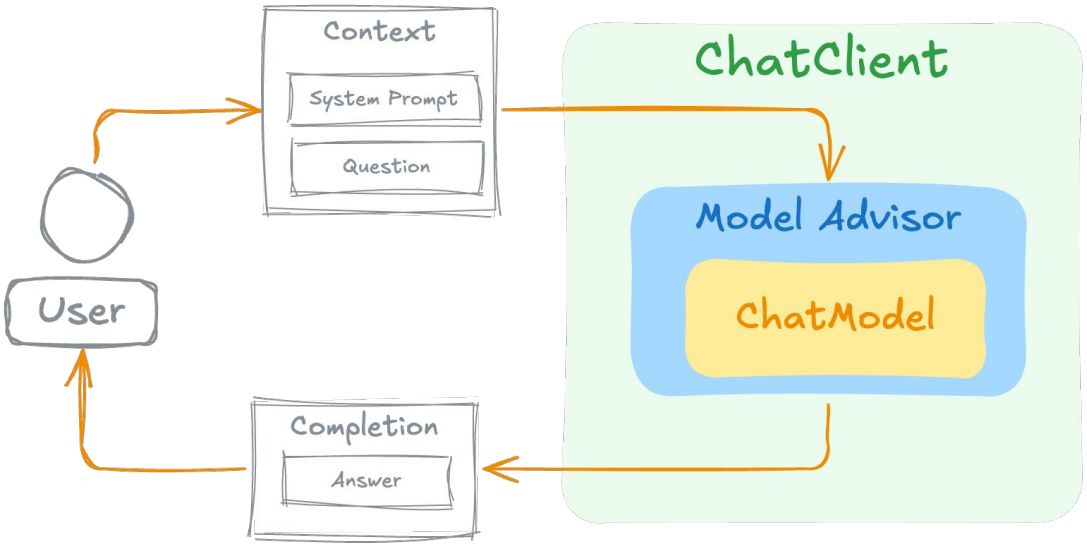
Spring AI Community

Agent Utils

- Agent Skills
- Sub-Agents
- Web Search Tool
- Web Fetch Tool
- File System Tools
- Shell Tools
- Auto-Memory/Dream
- ...

Agent To Agent (A2A)
Agent Client Protocol (ACP)
Spring AI AgentCore
Spring AI Harness Client

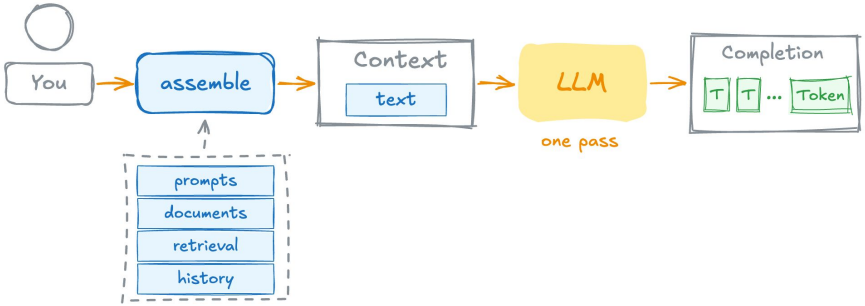
Relentless Context Curation



Two Ways to Manage Context

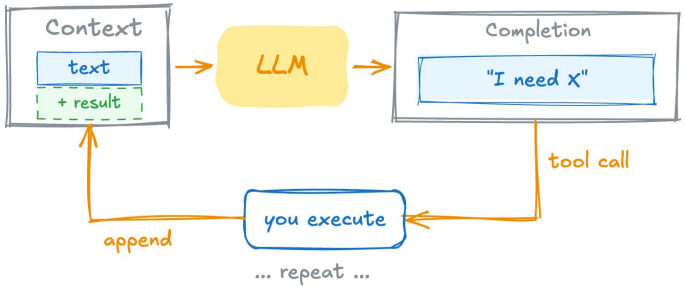
PRESCRIPTIVE

You curate the context



AGENTIC

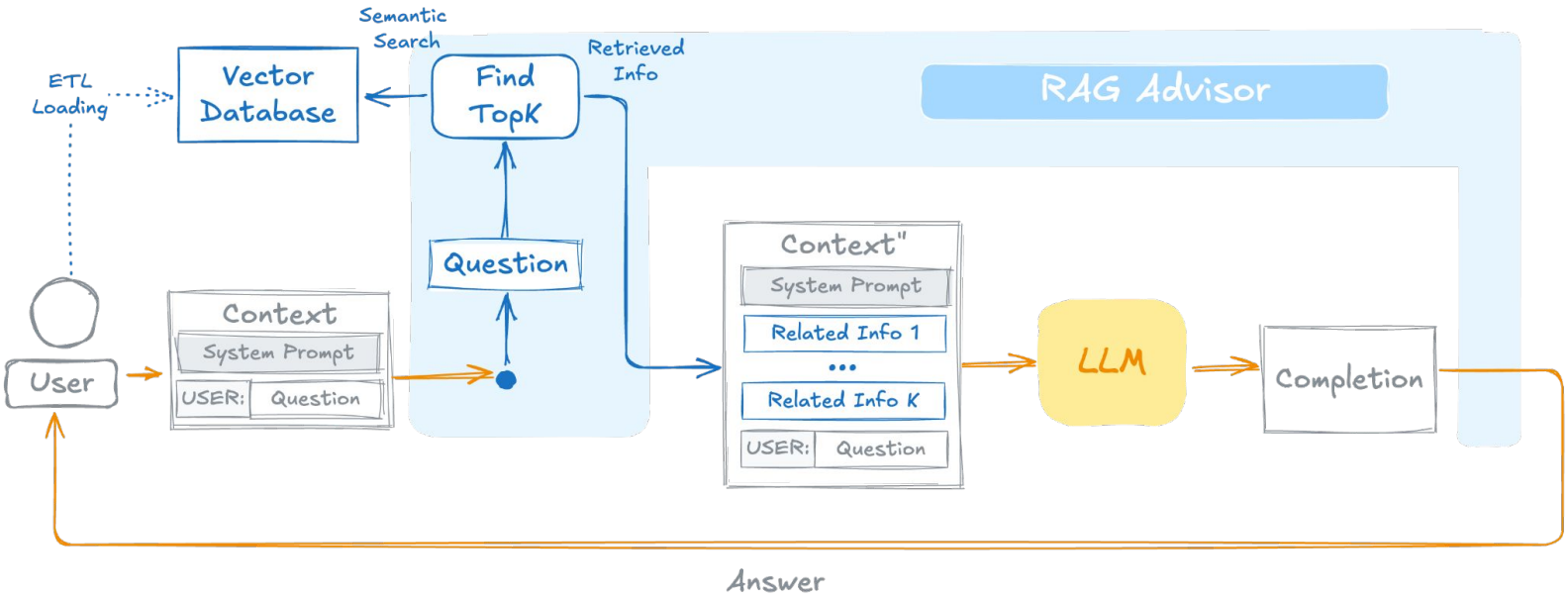
The model curates its context



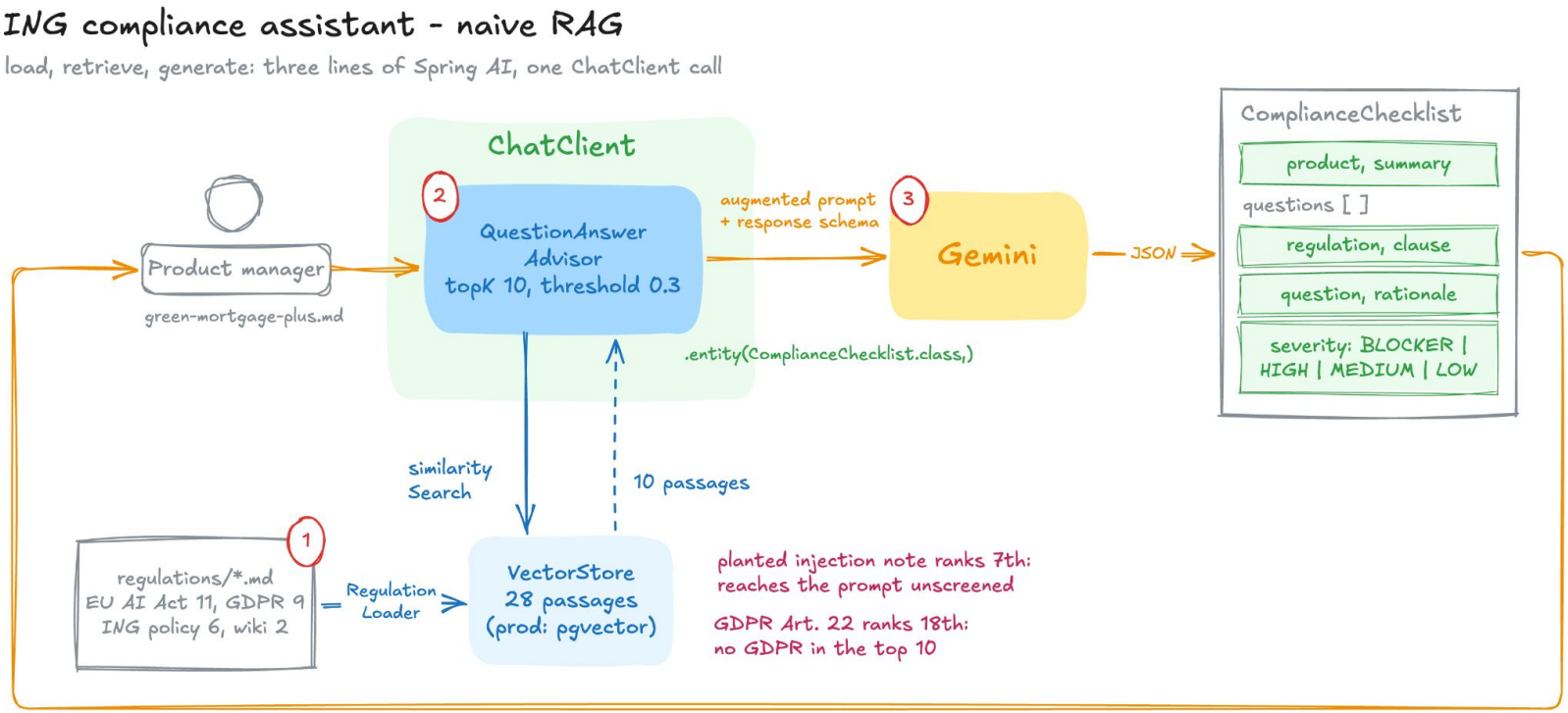
Hard problems need rounds of both.

When the model drives the rounds, that is the Agentic Loop.

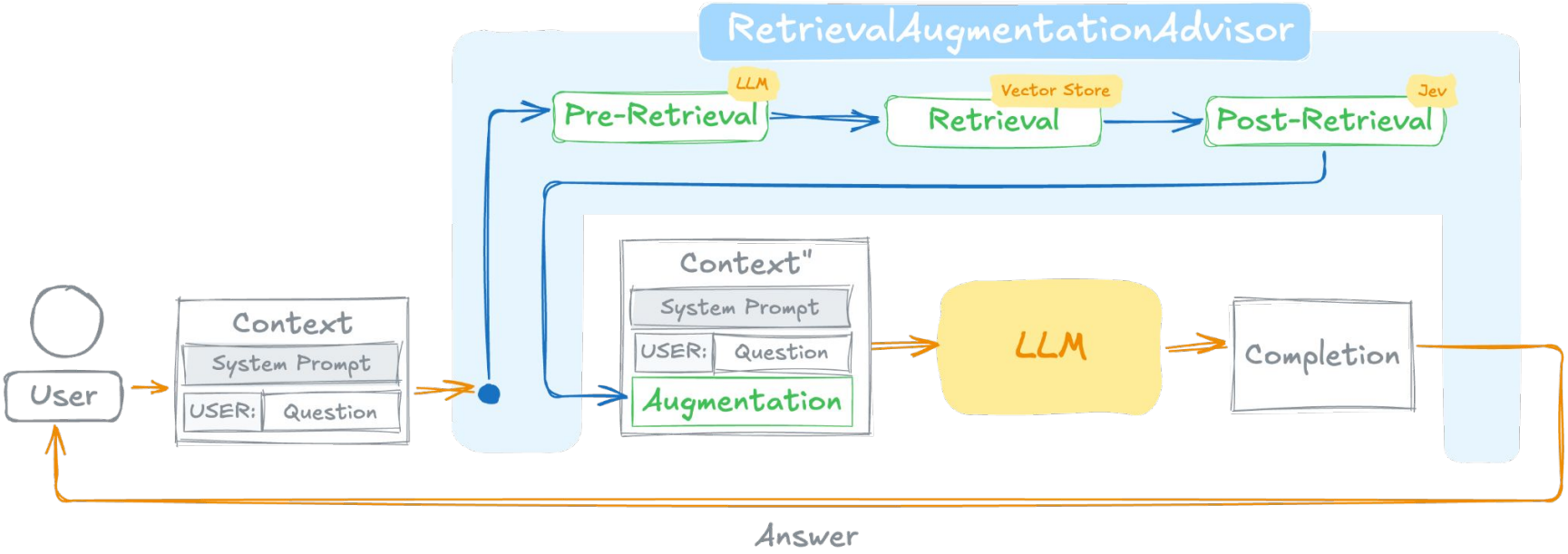
RAG Advisor


Semantic Search
ETL Loading
Vector Database
Find TopK
Retrieved Info
RAG Advisor
Question
Context"
System Prompt
Related Info 1
...
Related Info K
USER: Question
LLM
Completion
User
Context
System Prompt
USER: Question
Answer

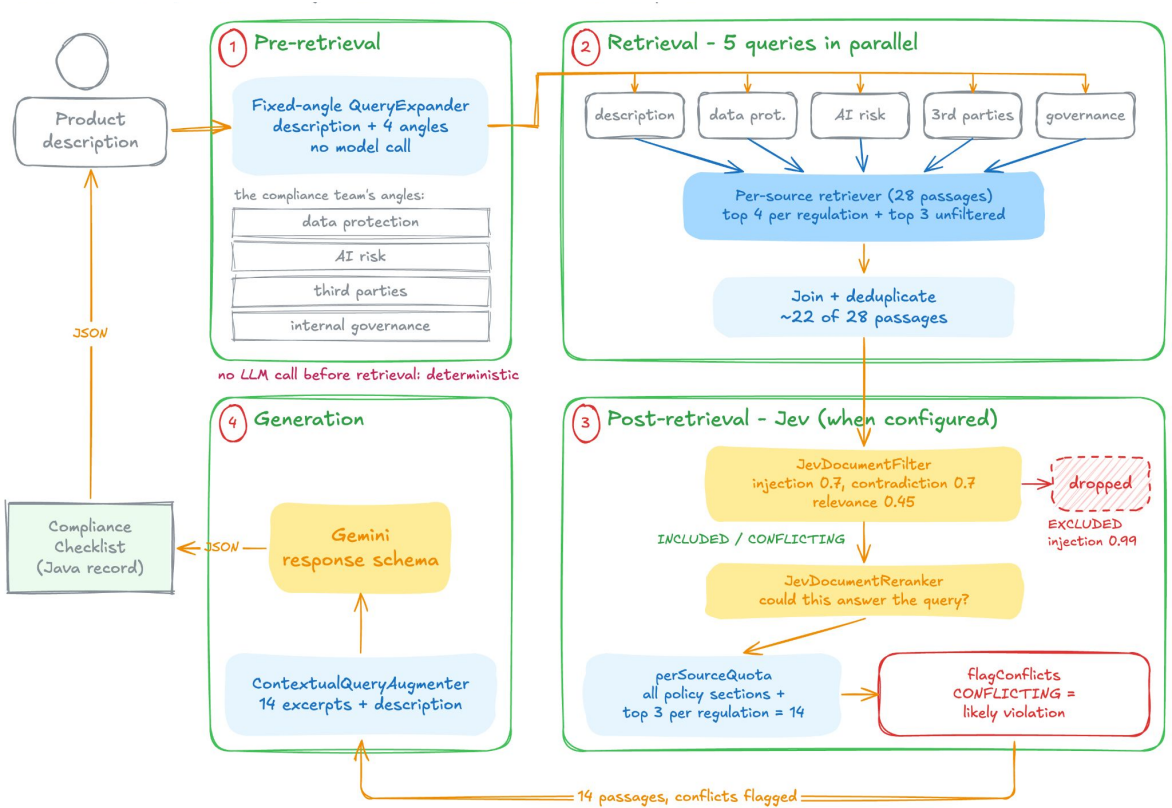
Naive RAG - Demo



Modular RAG + Jev - Advisor - Demo



Modular RAG + Jev



Naive vs Modular RAG + Jev

	Naive RAG	Modular RAG + Jev
Searches	1	5 (description + 4 regulatory angles)
Passages sent to Gemini	10	14
GDPR in the prompt	0	4: Art. 22, 44 to 46 and 35 as blockers, plus Art. 13 and 14
Planted Injection	in the prompt, rank 7	blocked by Jev (injection 0.99)
Compliance Questions (blockers)	9 (5)	14 (7)
Per request	16 s (gemini), 1 vector search	15 s (gemini), 5 vector searches, 44 Jev calls

Naive vs Modular RAG + Jev (2)

Searches: one search only finds passages that sound like the product description. The four angles search by regulatory topic.

Passages found, passages sent: the wider search finds more, and Jev screens and reranks them down to the 14 best.

GDPR in the prompt: for a product with automated credit refusals and a US data transfer, the naive search finds no GDPR at all.

Prompt injection: the planted intranet note reaches the naive prompt. Jev scored it 0.99 for injection and dropped it before Gemini saw it.

Compliance questions: the result is 14 questions instead of 9, four of them on GDPR.

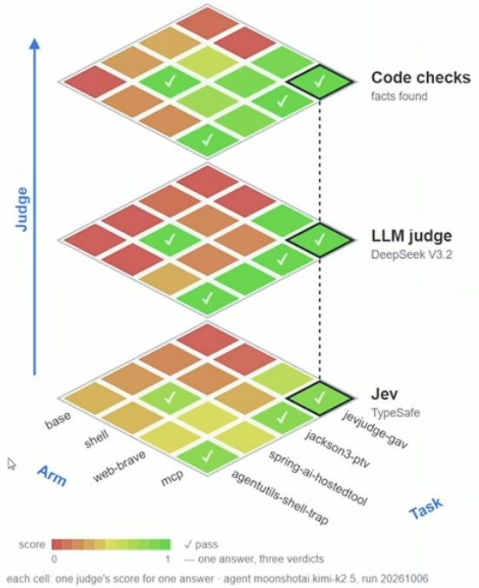
Request time: it's about the same. Searches and Jev add under 3 seconds, and the Gemini call varies more than that between runs.

Evals

Evals: Task, Arm, Judge

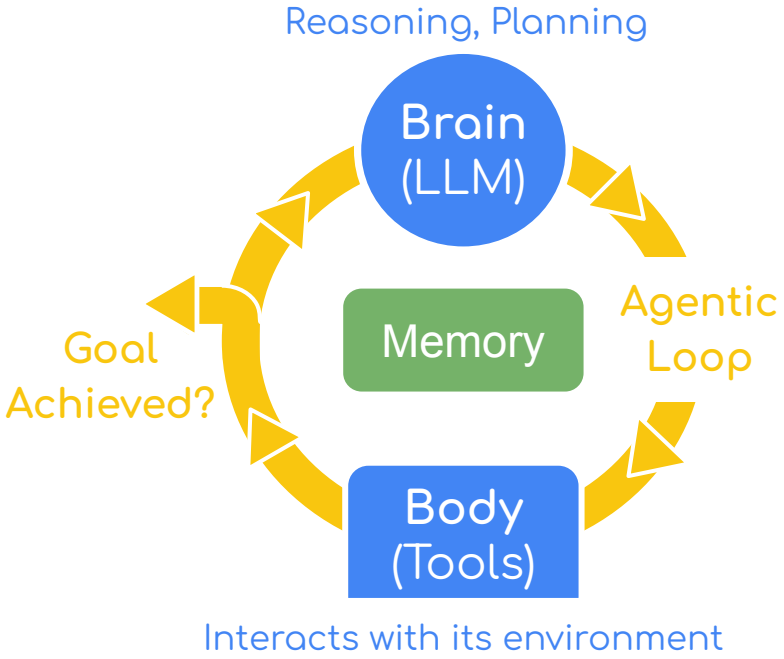
Tests for outputs of non-deterministic systems

1. **Task** What we are simulating
2. **Arm** Configuration of system to test
3. **Judge** How we determine success
4. **Metrics** How to measure accuracy & efficiency



Exquisite Evals for Skills & MCPs:
https://youtu.be/mebPFAKSpzw?si=lw96B31CZssC113p

An LLM agent runs tools in a loop to achieve a goal*

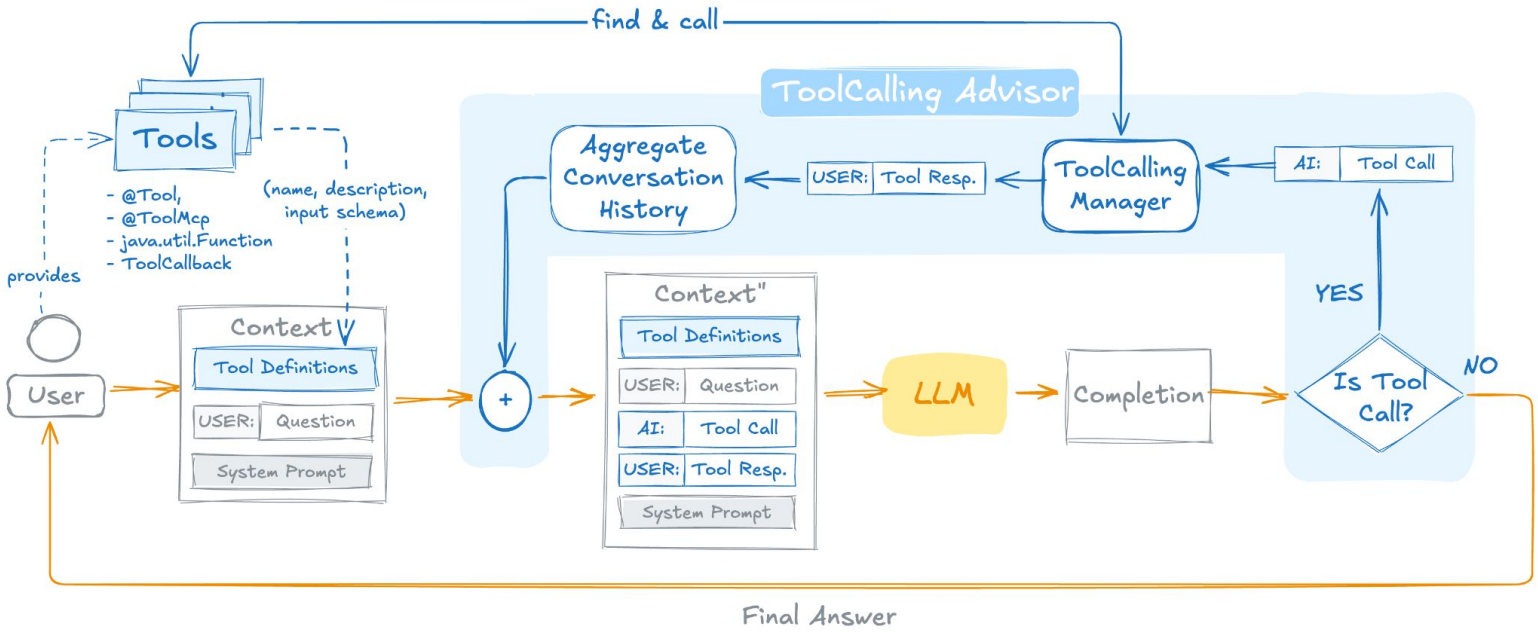


Leverages an AI models to interact with its environment in order to solve a user-defined task

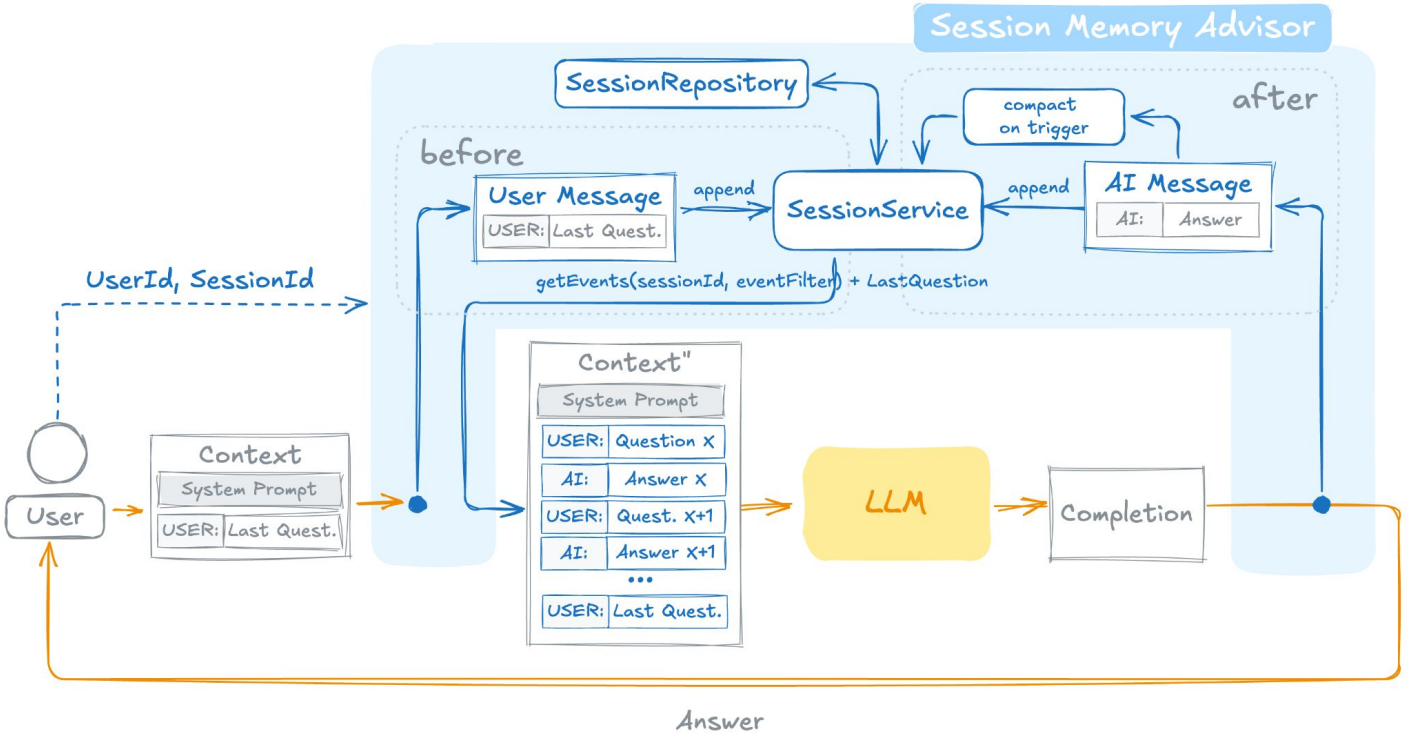
Relentless loop of assembling context, observing the result, and then re-assembling a context for the next step

(*) Simon Willison

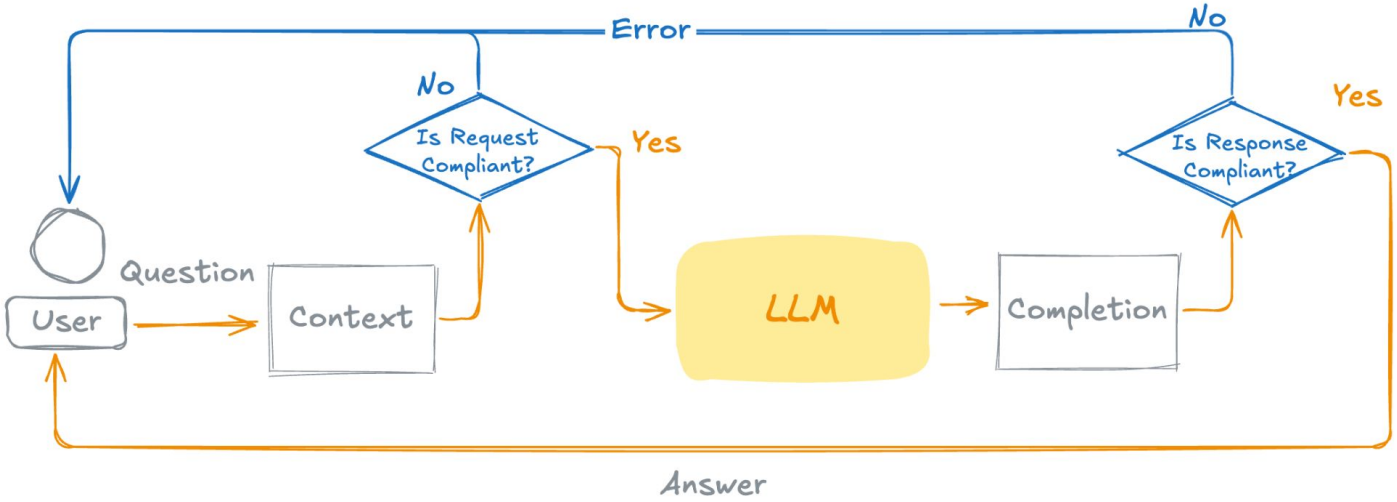
Autonomous LLM → Tool Calling Advisor



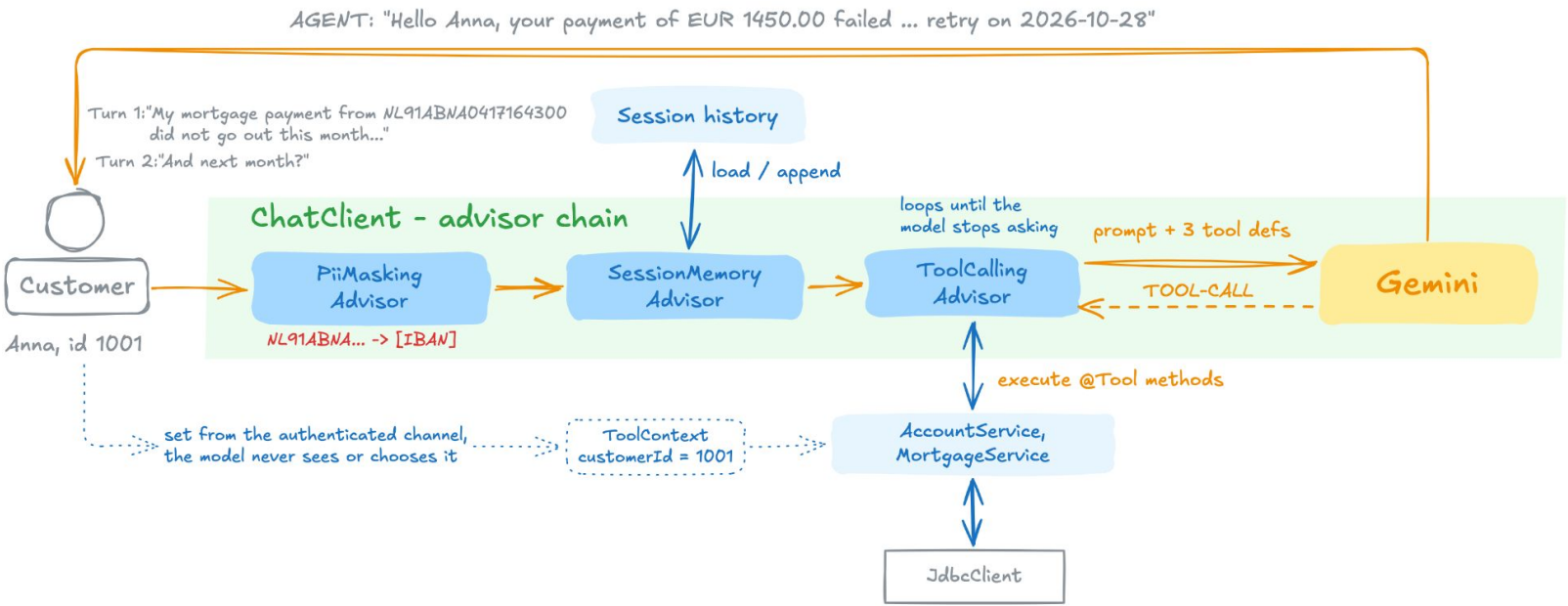
Stateless → Session Memory Advisor



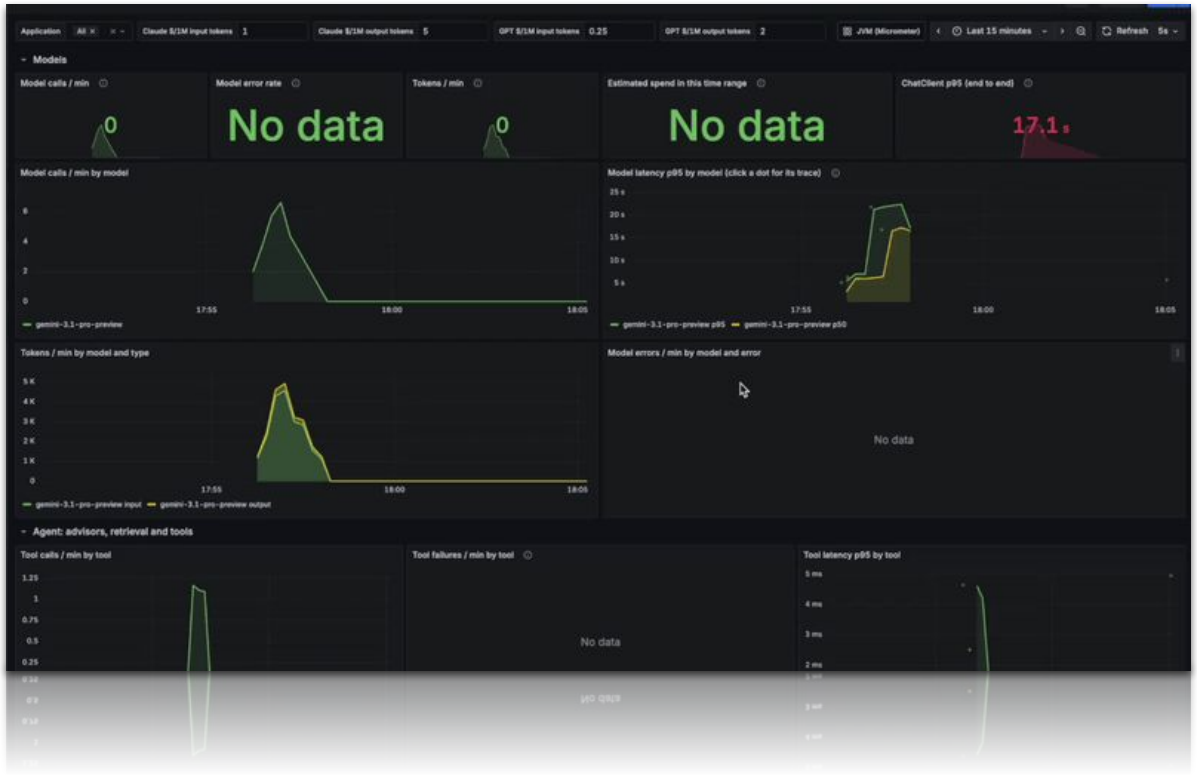
Hallucinations → Guardrails



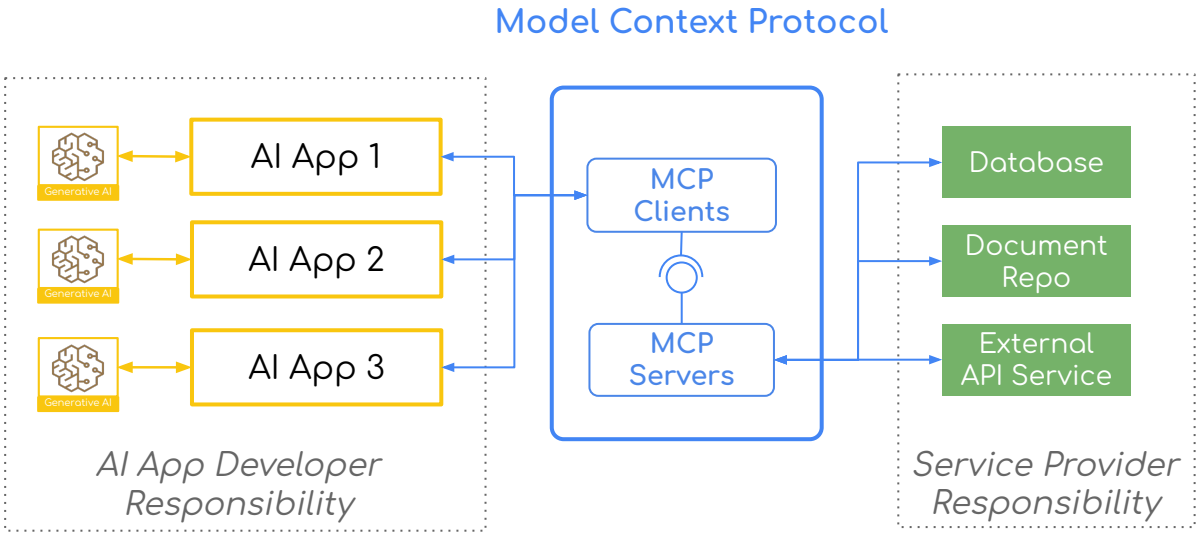
Customer Support Assistant - Demo



Observability



MCP AI Architecture



M + N integrations

References



Spring AI Reference



Spring AI Agent Utils



Agent Loops, Decoded Fundamentals to Agentic Patterns in Spring AI (Deep Dive)

Thank You

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